

José Ferreira

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EDUCATION

PhD in Physics <i>University of Aveiro</i>	09/2023 - PRESENT
MSc in Physics <i>Faculty of Sciences of the University of Lisbon</i>	09/2020 - 09/2022
BSc in Physics <i>Faculty of Sciences of the University of Lisbon</i>	09/2017 - 07/2020

RESEARCH INSTITUTES

Gr@v Gravitation group at the <i>Center for R&D in Mathematics and Applications</i> of the University of Aveiro, Portugal.	09/2023 - PRESENT
Institute of Astrophysics and Space Sciences Cosmology group at the <i>Institute of Astrophysics and Space Sciences</i> , Lisbon node, Portugal.	09/2021 - 07/2023

TEACHING

Unviersity of Aveiro Courses: Numerical and Statistical Methods, Physics for Earth Sciences	09/2025 - PRESENT
Faculty of Sciences of the University of Lisbon Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics	02/2022 - 07/2023

PUBLICATIONS

- J. Ferreira, G. Bozzola, C. Herdeiro, V. Paschalidis, M. Zilhão. “Electromagnetic duality degeneracy in dynamical black hole mergers”. (19 May 2026). [arxiv](#)
- J. Ferreira, C. Herdeiro, E. Radu, M. Zilhão. “Spontaneous spherical symmetry breaking of black holes with resonant hair”. *Phys. Rev. D* 114 (2 September 2026). [arxiv](#) [doi](#)
- J. Nicoules, J. Ferreira, C. Herdeiro, E. Radu, and M. Zilhão. “Splitting the Gravitational Atom: Instabilities of Black Holes with Synchronized/Resonant Hair”. *Phys. Rev. D* 113 (30 March 2026). [arxiv](#) [doi](#)
- J. Ferreira, T. Barreiro, J. Mimoso, and N. J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023). [arxiv](#) [doi](#)
- J. Ferreira, T. Barreiro, J. Mimoso, and N. J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022). [arxiv](#) [doi](#)

COMMUNICATIONS

TALKS

Electromagnetic Duality in Black Hole Mergers NewFunFICO Meeting, Frankfurt, Germany	15-17/06/2026 
(In)stability of Q-Hairy Black Holes NewFunFICO Meeting, Valência, Spain	20-22/05/2025 
Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry VI Amazonian Symposium on Physics, Belém, Brazil	18-22/11/2024  
Using Standard Sirens to test non-metric theories of gravity XVI Black Holes Workshop, Porto, Portugal	19-20/12/2023  



SEMINARS

Fate of Black Holes with Resonant Hair

Faculty of Sciences of the University of Lisbon, Portugal

12/12/2025



Cosmological Tests on Non-Metricity Based Theories of Gravity

University of Aveiro, Portugal

20/09/2023.



POSTERS

Splitting the (Gravitational) Atom

New Frontiers in Numerical Relativity, Mallorca, Spain

21-25/07/2025



Testing Λ -Free $f(Q)$ Cosmology

Dark Matter and Stars, Lisbon, Portugal

3-5/05/2023



PEER REVIEW

European Physical Journal

Reviewed 1 draft for European Physical Journal C.

02/2026 - PRESENT



DIGITAL SKILLS

High Level Languages Julia, Python, Mathematica

Multimedia Editing GIMP, Photoshop, Inkscape, Kdenlive, Audacity

Low Level Languages C, Fortran, CUDA

Web Development Quarto, Hugo

I am an avid Linux user, with experience using it on personal machines, servers, clusters and microcontrollers. I work mostly on a terminal based ecosystem, creating and making use of terminal utilities to develop my projects.



ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023

Institute of Astrophysics and Space Sciences

07/2023

Supervisor of two undergraduate projects on constraining modified gravity cosmological models with current and forecast datasets from multiple sources.

Podcast "Physically Speaking"



12/2020 - 12/2022

A student-driven podcast born during the pandemic, featuring interviews with faculty members about their lives both inside and outside academia.

Member of the Student Body for Physicists and Physical Engineers



10/2020 - 10/2021

Coordinator of the development of the digital tools of the student body of physicists and physical engineers.



OUTREACH

European Researchers' Night 2026

Coordinator of the participation of Gr@v in the European Researcher's Night in Aveiro, Portugal.

25/09/2026

Pint of Science 2026

Talk at the Pint of Science 2026 in Aveiro, Portugal.

20/05/2026



"Echoes and Shadows: How to Detect Black Holes"

A Presentation for high school students on what are black holes and how to find them.

03/05/2025

